Analytical Cumulative Examination

January 18, 1997

1. (20 points) The ionization constants for the diprotic acid oxalic acid {$(COOH)_2$} are $K_1 = 8.8 \times 10^{-2}$; and $K_2 = 5.1 \times 10^{-5}$. Oxalic acid has a distribution coefficient (K_D) between water and diethyl ether of 4.00. Calculate the distribution ratio (D) for oxalic acid between water and this organic solvent at pH 1.50.

2. (40 points) Two weak acids, HA ($pK_a = 4.5$) and HB (pK_a 6.2) have distribution coefficients (K_D defined as CCL_4/water) between CCL_4 and water of 3.0 and 3.5, respectively.

a. Assume that secondary equilibria are not available. Calculate the molar ratio of HB/HA remaining after 3 extractions.

b. Design a separation procedure for the removal of 98% of the HA from the aqueous solution, while retaining HB in the aqueous phase with no more than a 1% loss with each extraction.

3. (40 points) Recent discoveries in ion chromatography have led to the conception of single-column ion-exchange chromatography (IEC).

The ion exchange process can be represented by the equilibrium below:

Rs-E + X <====> Rs-X + E

where Rs represents the resin, E refers to eluent, and X is the sample ion. Rs-E can be thought of as the resin capacity, [cap].

a. Write the equilibrium constant, K_X for the process depicted above.

b. From a), derive the expression

$$\log k' = \log[cap] - \log[E] + \log(constant)$$

Hint: the distribution constant, K is given as

$$K = [X]_{phase1}/[X]_{phase2} = k' V_m/V_s$$

c. Comment on the significance of the equation derived in b) with respect to the development of single-column IEC.

d. Briefly describe the basis for suppressed (two column) ion-exchange chromatography.